Probabilistic Machine Learning

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Probabilistic Machine Learning
Advanced Topics

Kevin P. Murphy

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This book is dedicated to my wife Margaret,
who has been the love of my life for 20+ years.

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Preface

> I am writing a longer [book] than usual because there is not enough time to write a short one. (Blaise Pascal, paraphrased.)

This book is a sequel to [Mur22]. and provides a deeper dive into various topics in machine learning (ML). The previous book mostly focused on techniques for learning functions of the form $f : \mathcal{X} \to \mathcal{Y}$, where f is some nonlinear model, such as a deep neural network, $\mathcal{X}$ is the set of possible inputs (typically $\mathcal{X} = \mathbb{R}^D$), and $\mathcal{Y} = \{1, \ldots, C\}$ represents the set of labels for classification problems or $\mathcal{Y} = \mathbb{R}$ for regression problems. Judea Pearl, a well known AI researcher, has called this kind of ML a form of "glorified curve fitting" (quoted in [Har18]).

In this book, we expand the scope of ML to encompass more challenging problems. For example, we consider training and testing under different distributions; we consider generation of high dimensional outputs, such as images, text, and graphs, so the output space is, say, $\mathcal{Y} = \mathbb{R}^{256 \times 256}$; we discuss methods for discovering "insights" about data, based on latent variable models; and we discuss how to use probabilistic models for causal inference and decision making under uncertainty.

We assume the reader has some prior exposure to ML and other relevant mathematical topics (e.g., probability, statistics, linear algebra, optimization). This background material can be found in the prequel to this book, [Mur22], as well several other good books (e.g., [Lin+21b; DFO20]).

Python code (mostly in JAX) to reproduce nearly all of the figures can be found online. In particular, if a figure caption says "Generated by `gauss_plot_2d.ipynb`", then you can find the corresponding Jupyter notebook at probml.github.io/notebooks#gauss_plot_2d.ipynb. Clicking on the figure link in the pdf version of the book will take you to this list of notebooks. Clicking on the notebook link will open it inside Google Colab, which will let you easily reproduce the figure for yourself, and modify the underlying source code to gain a deeper understanding of the methods. (Colab gives you access to a free GPU, which is useful for some of the more computationally heavy demos.)

In addition to the online code, probml.github.io/supp contains some additional supplementary online content which was excluded from the main book for space reasons. For exercises (and solutions) related to the topics in this book, see [Gut22].

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About the cover

The cover illustrates a variational autoencoder (Chapter 21) being used to map from a 2d Gaussian to image space.

Kevin Patrick Murphy
Palo Alto, California
August 2023.

Changelog: see https://github.com/probml/pml2-book/issues?q=is%3Aissue+is%3Aclosed.

1 Introduction

> "Intelligence is not just about pattern recognition and function approximation. It's about modeling the world". — Josh Tenenbaum, NeurIPS 2021.

Much of current machine learning focuses on the task of mapping inputs to outputs (i.e., approximating functions of the form $f : \mathcal{X} \to \mathcal{Y}$), often using "**deep learning**" (see e.g., [LBH15; Sch14; Sej20; BLH21]). Judea Pearl, a well known AI researcher, has called this "glorified curve fitting" (quoted in [Har18]). This is a little unfair, since when $\mathcal{X}$ and/or $\mathcal{Y}$ are high-dimensional spaces — such as images, sentences, graphs, or sequences of decisions/actions — then the term "curve fitting" is rather misleading, since one-dimensional intuitions often do not work in higher-dimensional settings (see e.g., [BPL21a]). Nevertheless, the quote gets at what many feel is lacking in current attempts to "solve AI" using machine learning techniques, namely that they are too focused on prediction of observable patterns, and not focused enough on "understanding" the underlying *latent structure* behind these patterns.

Gaining a "deep understanding" of the structure behind the observed data is necessary for advancing science, as well as for certain applications, such as healthcare (see e.g., [DD22]), where identifying the *root causes* or mechanisms behind various diseases is the key to developing cures. In addition, such "deep understanding" is necessary in order to develop *robust* and *efficient* systems. By "robust" we mean methods that work well even if there are unexpected changes to the data distribution to which the system is applied, which is an important concern in many areas, such as robotics (see e.g., [Roy+21]). By "efficient" we generally mean data or statistically efficient, i.e., methods that can learn quickly from small amounts of data (cf., [Lu+21b]). This is important since data can be limited in some domains, such as healthcare and robotics, even though it is abundant in other domains, such as language and vision, due to the ability to scrape the internet. We are also interested in computationally efficient methods, although this is a secondary concern as computing power continues to grow. (We also note that this trend has been instrumental to much of the recent progress in AI, as noted in [Sut19].)

To develop robust and efficient systems, this book adopts a model-based approach, in which we try to learn *parsimonious representations* of the underlying "**data generating process**" (**DGP**) given samples from one or more datasets (c.f., [Lak+17; Win+19; Sch20; Ben+21a; Cun22; MTS22]). This is in fact similar to the scientific method, where we try to explain (features of) the observations by developing theories or models. One way to formalize this process is in terms of **Bayesian inference** applied to probabilistic models, as argued in [Jay03; Box80; GS13]. We discuss inference algorithms in detail in Part II of the book.[1] But before we get there, in Part I we cover some relevant background

1. Note that, in the deep learning community, the term "inference" means applying a function to some inputs to

material that will be needed. (This part can be skipped by readers who are already familiar with these basics.)

Once we have a set of inference methods in our toolbox (some of which may be as simple as computing a maximum likelihood estimate using an optimization method, such as stochastic gradient descent) we can turn our focus to discussing different kinds of models. The choice of model depends on our task, the kind and amount of data we have, and our metric(s) of success. We will broadly consider four main kinds of task: prediction (e.g., classification and regression), generation (e.g., of images or text), discovery (of "meaningful structure" in data), and control (optimal decision making). We give more details below.

In Part III, we discuss models for prediction. These models are conditional distributions of the form $p(\boldsymbol{y}|\boldsymbol{x})$, where $\boldsymbol{x} \in \mathcal{X}$ is some input (often high dimensional), and $\boldsymbol{y} \in \mathcal{Y}$ is the desired output (often low dimensional). In this part of the book, we assume there is one right answer that we want to predict, although we may be uncertain about it.

In Part IV, we discuss models for generation. These models are distributions of the form $p(\boldsymbol{x})$ or $p(\boldsymbol{x}|\boldsymbol{c})$, where $\boldsymbol{c}$ are optional conditioning inputs, and where there may be multiple valid outputs. For example, given a text prompt $\boldsymbol{c}$, we may want to generate a diverse set of images $\boldsymbol{x}$ that "match" the caption. Evaluating such models is harder than in the prediction setting, since it is less clear what the desired output should be.

In Part V, we discuss latent variable models, which are joint models of the form $p(\boldsymbol{z}, \boldsymbol{x}) = p(\boldsymbol{z})p(\boldsymbol{x}|\boldsymbol{z})$, where $\boldsymbol{z}$ is the hidden state and $\boldsymbol{x}$ are the observations that are assumed to be generated from $\boldsymbol{z}$. The goal is to compute $p(\boldsymbol{z}|\boldsymbol{x})$, in order to uncover some (hopefully meaningful/useful) underlying state or patterns in the observed data. We also consider methods for trying to discover patterns learned implicitly by predictive models of the form $p(\boldsymbol{y}|\boldsymbol{x})$, without relying on an explicit generative model of the data.

Finally, in Part VI, we discuss models and algorithms which can be used to make decisions under uncertainty. This naturally leads into the very important topic of causality, with which we close the book.

In view of the broad scope of the book, we cannot go into detail on every topic. However, we have attempted to cover all the basics. In some cases, we also provide a "deeper dive" into the research frontier (as of 2022). We hope that by bringing all these topics together, you will find it easier to make connections between all these seemingly disparate areas, and can thereby deepen your understanding of the field of machine learning.

compute the output. This is unrelated to Bayesian inference, which is concerned with the much harder task of inverting a function, and working backwards from observed outputs to possible hidden inputs (causes). The latter is more closely related to what the deep learning community calls "training".